Detection Range/Spatial Resolution Enhancement of Coherent Doppler Wind Lidar Based on Improved Simplex Coding

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***Abstract*— For the first time, a correlation-free coding technique is demonstrated for improving the detection range of coherent Doppler wind Lidar (CDWL) while roughly maintaining the spatial resolution of single-bit duration. Based on the probing pulses modulated by the improved simplex codes (ISCs), backscattered echoes can be considered as the linear superposition of the Lidar responses, each of which corresponds to different detection locations. Then, the Doppler spectra/wind profile approaching the spatial resolution of a single-bit duration can be recovered after decoding, time delay compensation, and accumulation. In contrast to those coding techniques that rely on correlation, the proposed method can be unaffected by the unequal amplitudes of coded bits that are stable between pulses. Theoretically, in simulated atmospheric echoes, it is confirmed that the degraded spatial resolution caused by wide overall emitting pulses can be mitigated while the reliable detection range can be improved. Additionally, experimentally constructed coding CDWLs using 9, 16, 32, and 64-bit ISCs show improvements in detection range that are consistent with the theory.**

***Index Terms*— Coherent Doppler wind Lidar (CDWL), detection range, simplex code (SC), spatial resolution.**

I. Introduction

PULSED coherent Doppler wind Lidar (CDWL) is widely used in weather forecast [1], air pollution monitoring [2], and aviation safety [3], etc. It detects the wind field by transmitting optical pulses into the atmosphere [4], receiving the coherent backscattered echoes [5] and extracting the Doppler frequency via power spectrum accumulation [6].

However, the dilemma between the spatial resolution and the maximum detection range remains a key constraint in the current CDWL [7]. Since CDWL calculates the periodogram by applying fast Fourier transformation (FFT) to contiguous echo samples, the minimum achievable spatial resolution is determined by the sum of the probing pulselength and the processed sampling length [8]. Limited by the nonlinear effect of laser fibers, the pulse peak power cannot be amplified infinitely [9]. Increasing the pulsewidth of the probing pulse improves the emitting pulse energy of the detected signal and accordingly enlarges the detection range but inevitably degrades the spatial resolution of the CDWL. Apart from that, using shorter pulsewidth and fewer sampling points would lead to a broadened spectrum [10], [11] and a poor spectral resolution [12], respectively.

A number of proposals and experimental demonstrations have been published to overcome this dilemma. The French Aerospace Lab tried to increase the maximum pulse peak power by using large-mode-area fiber and adding special stress to the erbium-doped fiber in 2015 [13]. In the same year, NASA shortened the pulse duration to 50 ns and achieved a minimum spatial resolution of 7.5 m [14]. These methods concentrated on using narrower pulse duration but cannot achieve an adequately long detection range. To address the spectral crosstalk brought by short probing pulses, Wolfe et al. provided a higher intermediate frequency (IF) of 1 GHz by adopting an electro-optic modulator (EOM) in CDWL [11]. Enlarging the IF is indeed helpful in distinguishing the baseband spectra and the signal, but cannot overcome the decrease in detection distance limited by emitting pulse energy.

Xia et al. [15], [16], [17], [18], [19], [20], pioneered a way to improve the spatial resolution of CDWL without decreasing the detection range by using correlation techniques which were conventionally used in Radars. They adopted complementary codes such as Golay codes, pseudorandom phase codes, phase shift keying, and continuous phase modulation, with a minimum bit duration of 12 ns, to achieve a spatial resolution of 1.8 m. In the correlation scheme, the backscattered signal should be correlated with the launched codes for decoding. If the amplified pulses are not highly consistent with the launched codes, serious sidelobes and crosstalk would be introduced [21], reducing or even destroying the effectiveness of the method. Thus, the signal generators, pulse modulators, and optical amplifiers are all supposed to be controlled precisely.

Simplex codes (SCs), which derive from Hadamard matrices, were applied in the optical time-domain reflectometer (OTDR) to improve its sensitivity and signal-to-noise ratio (SNR) as a correlation-free technique [22], [23]. The

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characteristic of the Hadamard matrix is that the product of the Hadamard matrix and its transpose is a unit square matrix [24]. To be specific, SCs can be obtained by removing the first row and the first column of the Hadamard matrix while changing the elements −1 and 1 in the Hadamard matrix to 1 and 0, respectively [24]. According to the theoretical predictions in [22], the sensitivity realized by SCs could be better than that of Golay codes. And in [23], an experimental OTDR system using 255-bit SCs showed a 9.2 dB SNR improvement. OTDR characterizes optical fibers by injecting optical probing pulses and detecting the backscattered optical signals, which possesses many common features with Lidar. However, we found that the SCs [24] cannot maximize the Lidar system's time and energy efficiency due to insufficiently emitted pulse bits.

In this study, a correlation-free coding Lidar based on improved SCs (ISCs) is proposed. The spatial resolution can be increased by decreasing the bit duration, while the detection range can be enlarged by increasing the code length. The original contributions of this coding method are as follows.

1) The ISC technique can fully realize the advantage of long emitting pulses by measuring the detection range of a single-bit duration multiple times. As a result, the resolution and SNR of the decomposed echo can both be enhanced. In contrast, a conventional noncoding Lidar adopting long probing pulses measures a wide detection range singly, which results in a blurred spatial resolution.
2) The ISCs coding Lidar can simplify the equipment requirements for practical applications compared to the existing coding techniques that require high correlation qualities between the launched pulses and the backscattered signals. Moreover, variations in the coded pulse amplitudes that remain constant between pulses, would not influence the measurement results of Doppler power spectra and wind profiles.
3) The ISCs, which are promoted by the authors for the first time, theoretically show higher efficiency in time and energy than the codes previously applied in the CDWL. Since the pulse sequences modulated by ISCs can launch more bits in identical measurement time, more energy can be combined in the decoding process, resulting in a larger potential SNR improvement.

The rest of the article is organized as follows. Section II gives the theoretical principle and the coding gain of ISCs. Sections III and IV show the performance of the ISCs coding technique when applied in the simulated atmosphere echoes and our self-developed Lidar system. A conclusion is finally drawn in Section V.

II. Principles

Firstly, Section II-A demonstrates an example of a coding Lidar using 4-bit ISCs to illustrate the working principle. Then the technique is generalized to arbitrary code length and the gains of different codes including the proposed ISCs, the Golay codes, and SCs are discussed in Section II-B.

A. Example of 4-Bit ISCs

Generally, the periodogram/power spectra to be estimated from the received echoes can be regarded as the Lidar's response toward the atmosphere via the probing optical pulse [25]. It is a function of the detection range z or the reception time t since there is a relationship of $z = ct/2$. It can be summarized as the convolution of the optical pulse waveform and the Lidar's backscatter impulse response [26]. However, due to the finite width of the input pulse, the resolution of the estimate of the periodogram degrades proportionately with the pulsewidth. Here, the periodogram/power spectrum of the wind Lidar at different reception time is defined as $P_\tau(t, f)$ when adopting a probing pulsewidth of τ. In the following sections of this article, $P_\tau(t, f)$ is abbreviated as $P_\tau(t)$.

Fig. 1 illustrates the composed backscattered echoes of the proposed coding Lidar using an example of 4-bit ISCs. In the time-range coordinates [27] in Fig. 1(a), the dotted lines with a slope c (speed of light) represent the propagation of probing pulse toward the atmosphere, while the dotted lines with a slope −c indicate the backscattering of laser from particles to the Lidar. At the time t_0, the Lidar receives the backscattered echoes that correspond to the leading edges of the first bit at $ct_0/2$ and the last bit at $c(t_0 - 3\tau)/2$. In other words, the received echo at each moment is the summation of the particle scattering signals ranging a pulsewidth, weighted according to the pulse waveform [28].

Assuming four sequentially spliced bits with duration τ together makes up the long probing pulse as in Fig. 1(a). Each bit of the probing signal generates a backscattering signal. It is their superposition that is detected and processed. In other words, the ISCs coding Lidar simultaneously measure many backscattered curves, each generated by an individual bit. Apparently, the bit sent out τ earlier would propagate $c\tau/2$ farther, proportionate to the time difference. Therefore, the raw processed periodogram at each moment is composed of four periodograms belonging to a single bit ($P_\tau(t)$), each with a different delay time according to the code sequences.

Then, if properly configuring the code sequences, it is feasible to recover the periodogram with the resolution of a single bit $P_\tau(t)$, just like handling a set of linear equations. Fig. 1(b) introduces a series of code sequences in the left part. As displayed in the right part, the backscattered echo of each modulated pulse (marked by the solid black line) is superposed by backscattered curves corresponding to the elements of code sequences (marked by colorful blocks). Combined with the code sequence listed in Fig. 1(b), if the bit duration is τ, the superposition process can be expressed as

$$\begin{aligned} S_1(t) &= \hat{P}_\tau(t-\tau) + \hat{P}_\tau(t-2\tau) + \hat{P}_\tau(t-3\tau) \\ S_2(t) &= \hat{P}_\tau(t) + \hat{P}_\tau(t-2\tau) + \hat{P}_\tau(t-3\tau) \\ S_3(t) &= \hat{P}_\tau(t) + \hat{P}_\tau(t-\tau) + \hat{P}_\tau(t-3\tau) \\ S_4(t) &= \hat{P}_\tau(t) + \hat{P}_\tau(t-\tau) + \hat{P}_\tau(t-2\tau) \end{aligned} \quad (1)$$

where $S_1(t)$, $S_2(t)$, $S_3(t)$, and $S_4(t)$ are the raw processed periodograms corresponding to the laser pulses modulated by the code sequences 0111, 1011, 1101, and 1110, respectively. $\hat{P}_\tau(t-\tau)$, $\hat{P}_\tau(t-2\tau)$, and $\hat{P}_\tau(t-3\tau)$ are the estimated periodograms of a single-bit duration τ with the time delays τ,

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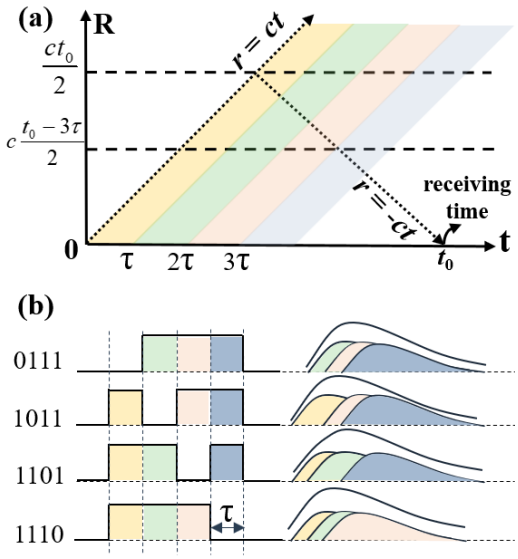


Fig. 1. Example of 4-bit ISCs coding Lidar. (a) Propagation and processing of coded pulses in the time-range diagram. (b) Code sequences and composed backscattered echoes.

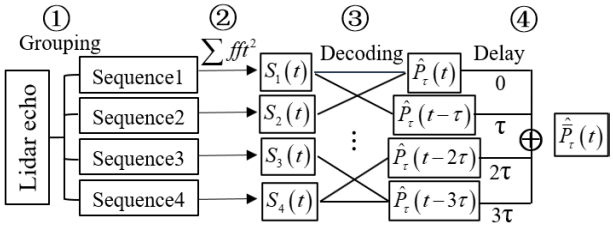


Fig. 2. Decoding process of the ISCs coding Lidar.

2τ, 3τ. Next, $\hat{P}_\tau(t)$ can be recovered from the composed periodograms by matrix multiplication as

$$\begin{bmatrix} \hat{P}_\tau(t) \\ \hat{P}_\tau(t-\tau) \\ \hat{P}_\tau(t-2\tau) \\ \hat{P}_\tau(t-3\tau) \end{bmatrix} = 1/3 \cdot \mathrm{G}_{\mathrm{decode}}(4) \cdot \begin{bmatrix} S_1(t) \\ S_2(t) \\ S_3(t) \\ S_4(t) \end{bmatrix} \tag{2}$$

where

$$G_{\mathrm{decode}}(4) = \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}.$$

The detailed proof of the decoding matrix can be found in Appendix. Finally, after recovering $\hat{P}_\tau(t)$ and its different time-delayed copies, adjusting the time differences and averaging all the copies yield the averaged resultant periodogram $\hat{\bar{P}}_\tau(t)$.

As shown by the example of 4-bit ISCs in Fig. 2, the entire processing operation of the ISC technology can be condensed into four steps: first, divide the backscattered echoes into groups according to the code sequences; next, calculate the corresponding periodogram of each echo group; then decode the composed periodograms by matrix multiplication; finally, apply time delay compensation and averaging.

Next, the theoretical SNR improvement value can be calculated based on the example of the 4-bit ISCs coding Lidar. Assuming the random noise in each measurement is white, with a variance of σ^2 which is equal to the average noise power [29]. Then the noise variance is $\hat{\sigma}^2_{\mathrm{averaging}} = (\sigma^2/L)$ if applying L pulses to conduct the conventional averaging method [25], [30]. Comparatively, utilizing the proposed 4-bit ISCs, the corresponding variance turns out to be $\hat{\sigma}^2_{\mathrm{coded}} = (5\sigma^2/9L)$. To be specific, the variance of the noise $e_x(t)$ in $S_x(t)$, $x = 1, 2, 3, 4$ is $(4\sigma^2/L)$, because there are $L/4$ pulses averaged in each group, as described in the following equation:

$$S_x(t) = \zeta_x(t) + e_x(t) \tag{3}$$

where $\zeta_x(t)$ is the periodogram of each sequence after removing the noise. During decoding in ③, combining (2),

$$\begin{aligned} \hat{P}_\tau(t) &= (S_2(t) + S_3(t) + S_4(t) - S_1(t))/3 \\ &= \frac{\zeta_2(t) + \zeta_3(t) + \zeta_4(t) - \zeta_1(t)}{3} \\ &\quad + \frac{e_2(t) + e_3(t)\, e_4(t) - e_1(t)}{3} \\ &= P_\tau(t) + (e_2(t) + e_3(t) + e_4(t) - e_1(t))/3. \end{aligned} \tag{4}$$

Since $\hat{P}_\tau(t)$ is 1/3 of the linear superpositions of all the four $S_x(t)$, $x = 1, 2, 3, 4$, the variance of the equivalent noise turns to $(5/9) \cdot (4\sigma^2/L)$ [22]. Because the four noises $e_x(t)$, $x = 1, 2, 3, 4$ contained in the numerator are generally uncorrelated, zero means random variables. For $\hat{P}_\tau(t - x\tau)$, $x = 0, 1, 2, 3$, the variance of the equivalent noise is the same. Finally in ④, the variance reduces to $(4/16) \cdot (5/9) \cdot (4\sigma^2/L)$, because there is a 4-time accumulation and 4-time averaging after aligning the delay.

B. Performance of Arbitrary Code Length

From the above example of 4-bit ISCs coding Lidar, it is clear that the coding gain results from replacing a long single pulse for multiple bits, amounting to measuring each high-resolution backscattered curve more times. In order to launch as many bits as possible and consume the least acquisition time for decoding, we design the ISCs which are more efficient than the SCs, as the following equation:

$$\begin{aligned} G_{\mathrm{code}}(M) &= \begin{bmatrix} 0 & 1 & \cdots & 1 & 1 \\ 1 & 0 & \cdots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & 0 & 1 \\ 1 & 1 & \cdots & 1 & 0 \end{bmatrix} \\ &= G_{\mathrm{decode}}(M) + \mathrm{diag}(2, \cdots, 2). \end{aligned} \tag{5}$$

If the code length is M, the ISCs can be defined by a $M * M$ square matrix G_{code}, where all the diagonal elements are 0 and the rest are 1. 0 and 1 indicate the off and on states of the laser. The corresponding decoding matrix G_{decode} can be easily obtained by replacing all the 0 with -1. If the code length is M, in the time it takes a conventional single-pulse Lidar to take L measurements, the ISCs coding scheme can effectively make $L(M-1)$ measurements. Comparatively, the SCs contain 1 in $(M+1)/2$ places in each row, corresponding to making $L(M+1)/2$ measurements of single-bit's backscattered curve in identical measurement time.

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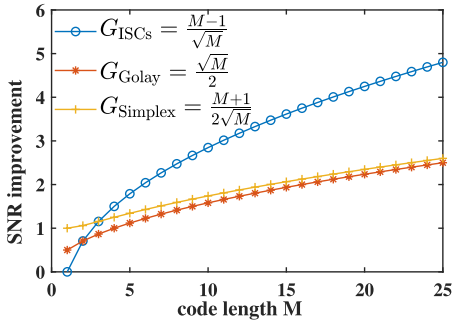


Fig. 3. SNR improvement of different codes.

Another advantage is that the amplitude deviation of different bits inside each row does not influence the effect of the ISCs coding method if the deviation is constant between pulses. If the intensities of the code sequences change to $[0AB \cdots XY]$, after modifying elements in G_{decode} accordingly, $\hat{P}_\tau(t)$ can still be recovered. For example, if the first code sequence of the 4-bit ISCs changes to $0ABC$, then the first equation in (1) can be modified as

$$S_1(t) = A\hat{P}_\tau(t-\tau) + B\hat{P}_\tau(t-2\tau) + C\hat{P}_\tau(t-3\tau). \quad (6)$$

Then $\hat{P}_\tau(t)$ and its time-delayed copies can still be recovered by jointly solving all the equations in (1). Although the amplitude inside each bit is supposed to be steady, it is easy to accomplish when the bit duration is short. Overall, it means that the amplified pulse bits can be nonidentical based on this correlation-free coding technique if they were determined before decoding.

If the code length is M and the total number of measured pulses is L, the variance of ISCs coding is

$$\hat{\sigma}^2_{\text{coded}} = \frac{\sum_M \sum_M \frac{M\sigma^2}{L}}{M^2 \cdot (M-1)^2} = \frac{M\sigma^2}{(M-1)^2 L}. \quad (7)$$

The derivation way is similar to the example shown in Section II-A from (3) to (4): the variance of the periodogram $S_x(t)$ of each code sequence is $(M\sigma^2/L)$; then, in the denominator, M-1 times averaging are introduced in the decoding process in ③ and another M-times averaging is required in ④ after the time delay adjustment; in the numerator, a number of $M^2 S_x(t)$ are accumulated in total, in which the measurement noises are uncorrelated from sample to sample, contributing to M^2 times of the noise power.

Compared with the Golay coding, the simplex coding, and the conventional averaging method, the ISCs coding is superior in promoting the SNR. With a code length of M and total measurement pulses of L, the average mean square errors of Golay codes, SCs, and conventional averaging are $(4\sigma^2/ML)$ [31], $(4M\sigma^2/(M+1)^2 L)$ [22], and (σ^2/L) [25], respectively. Theoretically, the square of SNR is proportionate to the reciprocal of the normalized noise power [21]. Based on this, Fig. 3 plots the SNR improvements (G_{ISCs}, G_{Golay}, G_{Simplex}) of the three coding techniques versus the conventional averaging scheme with identical measurement time. The ISCs perform better than the Golay codes and the SCs since $M > 2$ and $M > 3$, respectively. And the SNR gap gets wider as the code length becomes longer. Noteworthy is that, in the ISCs coding, there is no gain when $M \leq 2$ because no additional energy is emitted compared to the noncoding condition. Similarly, there is no SNR promotion when $M < 4$ for Golay codes.

According to the Lidar equation [8], [9], the detection range R is inversely proportional to $\sqrt{\text{SNR}}$. Therefore the detection range of ISCs coding Lidar can be extended as $R_{\text{coded}} \sim (\sqrt[2]{M-1}/\sqrt[4]{M}) R_{\text{averaged}}$, where R_{averaged} denotes the detection range using the conventional averaging method with identical processing pulses. The SNR improvement/noise reduction would finally broaden Lidar's detection range, providing a visible way to validate the ISCs coding technique in the following simulations and experiments.

III. Simulations

Prior to the realization of experimental tests, a noncoding CDWL and an ISCs coding CDWL are established in simulation to compare their theoretical performances.

A. Echoes of Noncoding CDWL and ISCs Coding CDWL in Time Domain

Following the signal path from the transmitter to the receiver and combining with the classical model of atmospheric scattering [8], [9], the normalized coherent backscattered echo can be expressed by the following equations:

$$Z(t) = \sum_{i=1}^{N_p} \left\{ \frac{T(R_i)}{R_i^2} \exp\left[\frac{4\pi j R_i(t)}{\lambda}\right] Y(R_i) + I_{\text{n}} \right\} \quad (8)$$

$$I_n = I_{R_0} \cdot 10^{-\frac{\chi_{\text{SNR}}}{20}} \cdot \left[\text{randn}\left(1, N_p\right) + j \cdot \text{randn}\left(1, N_p\right)\right] \quad (9)$$

$$T(R_i) = \exp\left[-\int_0^{R_i} \varepsilon(r) dr\right] \quad (10)$$

where $Z(t)$ is the normalized coherent echo containing the random noise I_n obtained in each measurement, $N_p = 10^6$ is the number of particles involved by the probing pulse in a single sample, $\lambda = 1550$ nm is the laser wavelength. Given that only the relative values of the signal and noise are useful in the simulation, so constant parameters related to the system hardware can be ignored by setting to 1 in (8). In (8), the exponent term $4\pi j R_i(t)/\lambda$ reflects the instantaneous phase caused by the motion of the ith particle located at R_i, from which the Doppler frequency shift can be derived. The term $Y(R_i)$ corresponds to the emitted pulse shape. For the conventional CDWL and the ISCs coding CDWL, $Y(R_i)$ can be set to a square pulse or the encoded sequence that follows (5), respectively. As shown in (9), the complex random noise I_{n} is set by dividing the intensity of the echo signal in the range R_0 and a given SNR χ_{SNR}. Then, in direct proportion to the square of the detection range, the SNR of the Lidar's echo falls. The atmospheric transmission introduced in (10) describes the absorption and scattering by air molecules and aerosol particles along the propagation path, where $\varepsilon(r) = 2 \times 10^{-4}$ m^{-1} is the radiation extinction coefficient.

Next, echoes of a 4-bit ISCs coding Lidar with a 100 ns bit duration are derived and processed to verify if the decoding

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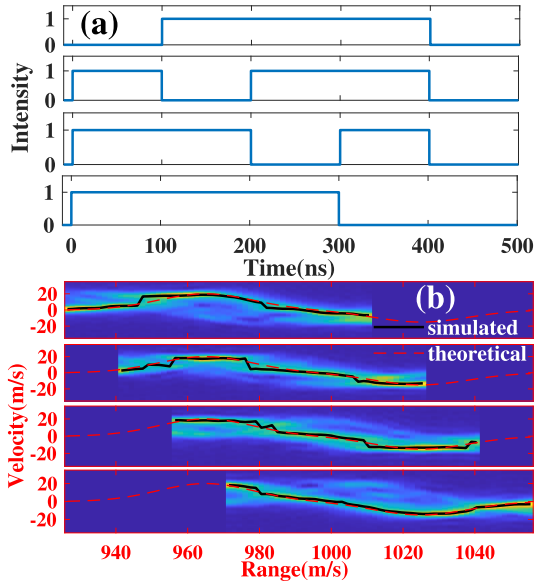


Fig. 4. Transmitting pulses and corresponding decoding power spectra of a simulated 4-bit ISCs Lidar. (a) Coded transmitting pulse sequences. (b) Time-delayed power spectra obtained by decoding.

performance meets the expectations in Section II-A. A fast-varying wind field consisting of two Gaussian-shaped profiles is built up, centering in 1000 m with a range of 100 m as

$$V_{\text{short}} = 20\exp\left[-\frac{(R-960)^2}{350}\right] - 15\exp\left[-\frac{(R-1025)^2}{350}\right]. \tag{11}$$

Based on (11), the wind field rapidly changes from −15 to 20 m/s within a short range of 100 m, which is convenient for observing the time-delayed power spectra obtained by decoding. If the estimated velocities derived from the calculated time-delayed power spectra agree well with the preset, the decoding procedure may be correct, and the ISC technique can be seen as a feasible means in wind Lidar.

The simulated 4-bit ISCs coding Lidar's pulse profile is shown in Fig. 4(a) and is utilized as the $Y(R_i)$ in (8) to derive the backscattered echoes. The sampling rate in the simulation is 100 MHz corresponding to a 1.5-m sampling interval. Meanwhile, the sampling points adopted for calculating the power spectrum is 6, resulting in a range bin length ($(Tp + 6Ts)c/2$) of 15 m for 40 ns pulsewidth. The pulse repetition frequency (PRF) is 10 KHz. In order to reduce computational effort, the number of pulses for spectrum incoherent accumulation is 90, which seems much smaller than the value commonly used in practical applications but enough for doing comparisons in simulation.

Following the steps summarized in Fig. 2, after dividing the Lidar echoes into four groups and calculating the averaged power spectrum of each group, the periodograms $\hat{P}_\tau(t) \sim \hat{P}_\tau(t-3\tau)$ can be decoded by matrix multiplication [see (2)], as displayed in Fig. 4(b). Seen from the x-coordinate, each column contains one power spectrum relating to one distance/time. The vertical coordinate indicates the velocity ranging from −35 to 35 m/s. The resulting four periodograms delay each other by a range corresponding to the bit duration of 100 ns. Additionally, it is clear that their estimated velocity profiles, marked by the black solid lines, are consistent with the preset background velocities marked by the red dashed lines. It can be deduced that a more precise estimate can be retrieved from $\hat{\hat{P}}_\tau(t)$ after averaging all the delayed copies. Therefore, the decoding process of applying ISCs coding in CDWL is implementable. The improvements in detection range/spatial resolution are carefully analyzed in Sections III-B and III-C ,respectively.

B. Comparison of Detection Ranges of Noncoding Lidar and 9-Bit ISCs Coding Lidar

In order to explore the detection range promotion brought by ISCs coding, a large-scale fluctuating wind field lasting for more than 1 Km in the range of −6–6 m/s is considered in this simulation, which is described as

$$V_{\text{long}} = 3.383 \cdot \sin(aR - 2.736) + 2.637 \cdot \sin(bR + 1.713) \tag{12}$$

where $a = 0.006147$, $b = 0.01621$.

Combined with the performance analysis in Section II-B, the coding gain enlarges with the code length. To observe a more obvious difference between coding and noncoding Lidar, the pulse sequences of 9 bits and 16 bits with a 40 ns bit duration are adopted in this section. Comparatively, the detection results of noncoding Lidars with 40 and 360 ns pulsewidth are given in Fig. 5(a) and (b). The red solid lines depict the theoretical velocity profile while the dark dashed lines mark the estimated velocities according to the peak of the power spectra. For the four Lidars in Fig. 5, the SNR(χ_{SNR}) at the start point of 200 m(R_0) is 5 dB and the peak power of all the probing pulses is set the same.

Apparently, the noncoding Lidar with 40 ns pulsewidth possesses the most fluctuating estimated velocity profile due to the heavily contaminated power spectra background. Along with the increase in the detection range, the nominated velocities jump up and down, which are far from the preset values. At this time, the estimated results become unreasonable. When the pulsewidth is increased to 360 ns, the reliable detection range of the noncoding Lidar extended from less than 600 m to about 800 m. As seen from Fig. 5(c), the detection range of the ISCs coding Lidar is comparable with that of the noncoding Lidar with 360 ns pulsewidth, but the velocity error seems smaller in detail. And the background power spectra appear cleaner with less interfering noise. This indicates that the ISCs coding Lidar proposed in this manuscript can have a comparable reliable detection range to the noncoding Lidar with the same overall pulsewidth. Additionally, the detection range of 9-bit coding Lidar is about 1.4 times that of noncoding Lidar, which is basically consistent with the range improvement derived in Section II-B.

Moreover, as seen from Fig. 5(d), when the code length increases from 9 to 16, the detection range continues to extend to around 1100 m. Though the enlargement of the detection range is consistent with the theory, the deviation of the estimated wind profile toward the preset seems to become larger. This may be because the information of adjacent

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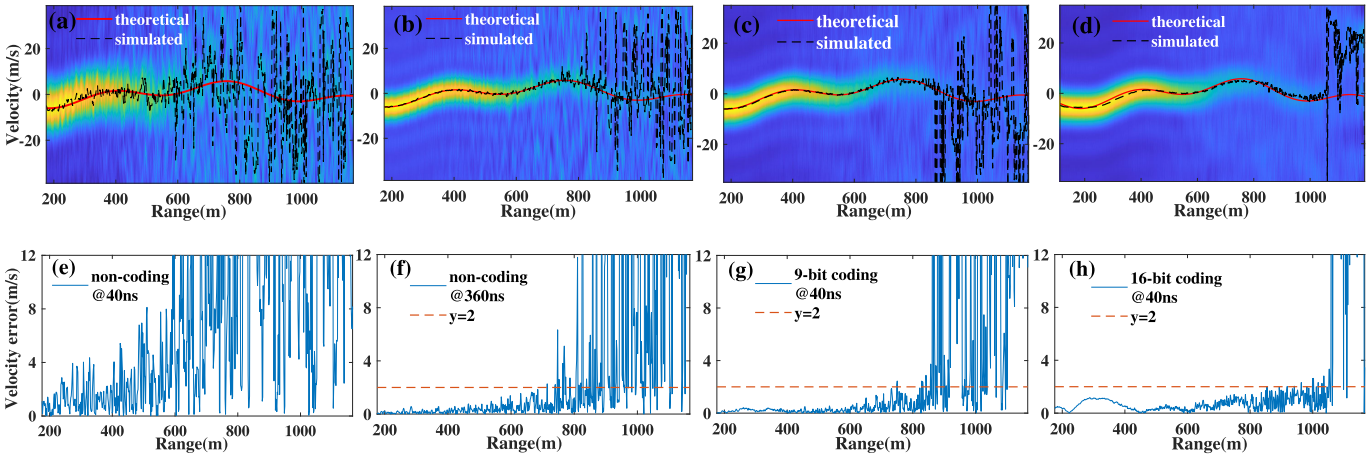


Fig. 5. Simulated spectra, corresponding estimated velocities and velocity errors according to the peak of the power spectra in a large-scale wind field. Results of the noncoding Lidar with (a) and (e) 40 ns pulsewidth and (b) and (f) 360 ns pulsewidth. Results of (c) and (g) 9-bit and (d) and (h) 16-bit ISCs coding Lidar with 40 ns bit duration.

codes degrades the resolution of the estimated periodograms $\hat{P}_\tau(t - x\tau), x = 0, 1, \ldots, 15$. Therefore, in order to achieve a higher consistency of the estimated and desired wind profiles, the overall amount of coding time should not be too long. Or the value of y in the decoding matrix should be changed to suit the code length at this time.

C. Comparison of Spatial Resolution of Noncoding Lidar and 9-Bit ISCs Coding Lidar

This section aims at comparing the detection accuracy between the coding and noncoding Lidars. It is suitable to use the wind field determined by (11) which described a fast-varying velocity profile in a small range scale. To investigate how the pulsewidth and the coding technique affect the estimation accuracy, no noise is added to the Lidar echo. Other parameters are the same as in Sections III-A and III-B.

Similarly, Fig. 6 gives the averaged periodograms $\hat{P}_\tau(t)$ and corresponding estimated velocities of noncoding Lidars with 40 and 360 ns pulsewidths as well as 9-bit ISCs coding Lidars with 40 and 10 ns bit durations. Among the first three Lidars, the noncoding Lidar with a pulsewidth of 40 ns has the least mean velocity error while the noncoding Lidar with a 360 ns pulsewidth is the most inaccurate, which is in line with the theory. The average error and the correlation coefficient of the 9-bit ISCs coding Lidar with 40 ns bit duration are in the middle of the first and the second Lidar.

The ISCs coding Lidar's velocity estimation accuracy is indeed higher than that of the noncoding Lidar with the same overall pulsewidth. However, it fails to achieve the same level of resolution as the noncoding Lidar adopting the same single-bit duration. The reason for the less-than-expected resolution here may be: firstly, the decoding matrix used here contains information from adjacent bits, which limits the resolution in principle; secondly, there may be a deviation between the true frequency and the frequency obtained by the power spectrum estimation using a finite number of sampling points due to the overlapping of processing area in FFT operation.

As shown in Fig. 6(d), when the bit duration is decreased to 10 ns, the average measurement error and the correlation coefficient are greatly promoted. In detail, the estimated velocities in the middle of the detection area closely match the preset values, however, those in the two ends are slightly misfit. This happens because the averaging times at both ends are less than those in the middle resulting from the accumulation process of time-delayed periodograms $\hat{P}_\tau(t - x\tau), x = 0, 1, \ldots, 8$. For ease of viewing, only the middle part is remained by keeping the x-axis of different plots the same.

To sum up, it is proved in simulation that the ISCs coding can partially mitigate the degradation of spatial resolution while achieving a similar effective detection range compared with the Lidar adopting identical overall pulsewidth. In addition, the spatial resolution can be increased by decreasing the bit duration, while the detection range can be enlarged by increasing the code length. However, since the decoding matrix retains some information from adjacent bits to eliminate the interference of noise, the code length can also influence the resolution. Therefore, the code length is not supposed to be too large except for changing y in the decoding matrix. The simulation results of the ISC technique are essentially consistent with the theory, showing great application potential.

IV. Experiments

To practically verify the SNR/detection range enhancement in the proposed ISCs coding Lidar, we experimentally construct a wind Lidar which can modulate the probing pulses according to the given code sequences.

A. Experimental Setup

The experimental setup, as shown in Fig. 7, consists of a narrow-linewidth single-frequency laser (SFL) emitting continuous light in 1550 nm, a 1% optical coupler (OC1) splitting the laser into one signal branch (99%) and the other reference branch (1%). In the signal branch, the Mach–Zehnder modulator (MZM) receives a microwave code waveform from the arbitrary waveform generator (AWG) and chops the laser into coded pulses by carrier suppression modulation. An acousto-optic modulator (AOM) is used to gate the MZM pulse

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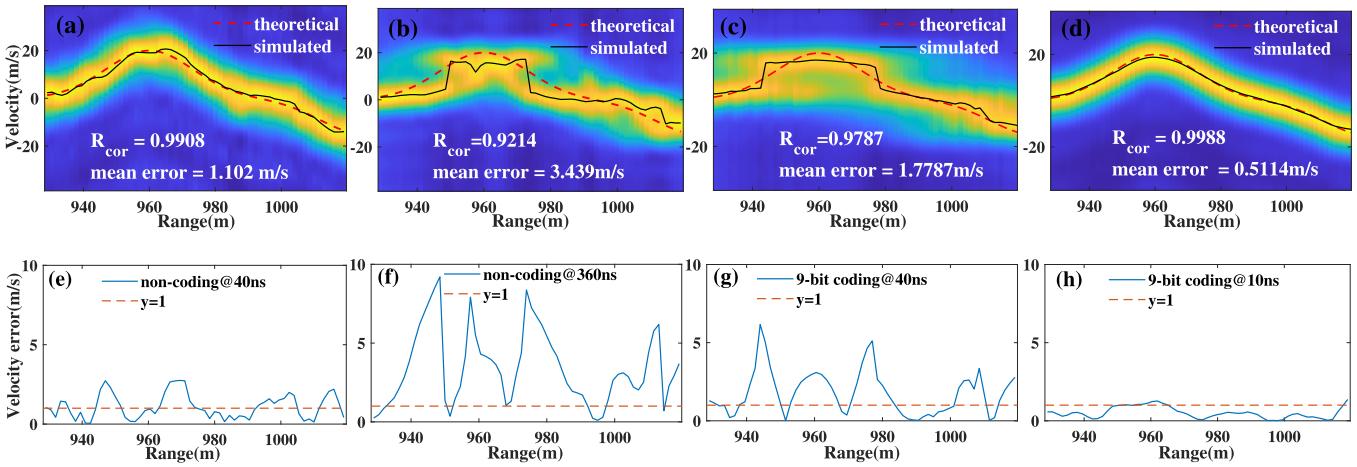


Fig. 6. Simulated spectra, corresponding estimated velocities and velocity errors according to the peak of the power spectra in noise-free conditions. Results of the noncoding Lidar with (a) and (e) 40 ns pulsewidth and (b) and (f) 360 ns pulsewidth. Results of the 9-bit ISCs coding Lidar with (c) and (g) 40 ns bit duration and (d) and (h) 10 ns bit duration.

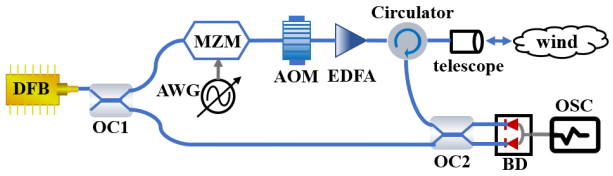


Fig. 7. Schematic configuration of the ISCs coding Lidar. SFL. OC: optical coupler. MZM. AWG. AOM. EDFA. BD. OSC.

TABLE I
PARAMETERS OF THE ISCS CODING LIDAR

Parameters	Value
pulse repetition frequency	10 KHz
sample rate	1 GSps
temporal resolution	0.5 s
reference power	0.4 mW
range bin	15 m
AOM frequency shift	200 MHz
output power(9-bit@40ns and non-coding@360ns)	0.99 W
output power(non-coding@40ns)	0.32 W
output power(16-bit@10ns)	0.48 W
output power(32-bit@10ns)	0.98 W
output power(64-bit@10ns)	2 W

sequence and improve the pulse suppression ratio. An erbium-doped fiber amplifier (EDFA) is placed after the AOM to enlarge the output pulse energy. After that, the coded pulses are sent out through a collimating telescope and received by a circulator. Finally, a 50% optical coupler (OC2) combines the received echo and the reference beam on a balanced detector (BD) that is sampled by an oscilloscope (OSC).

However, since the wind field in the author's environment is generally flat, and there is no alternative wind speed measurement equipment with a higher resolution to provide a reference wind speed, it is difficult to compare the velocity inaccuracy of a fast-changing wind field as simulated in Section III-C. As a result, the experiment compares the achievable detection range as a reflection of the improvement in SNR, in order to assess the viability of the ISC technique.

B. Results From Noncoding Lidars and 9-Bit Coding Lidar With 40 ns Bit Duration

Firstly, a 9-bit ISCs coding Lidar with a 40 ns bit duration is constructed. For comparison, noncoding Lidars with 40 and 360 ns pulsewidths are configured using the same system shown in Fig. 7. The key system parameters are summarized in Table I. At a sampling rate of 1 Gsps, the raw power spectrum is the summation of 5000 pulse echoes that are FFT processed with a range bin of 15 m. The accumulation of 5000 shots combined with the 10 KHz PRF yields a temporal resolution of 0.5 s.

The average output power of the 9-bit coding Lidar and the noncoding Lidar with 360 ns pulsewidth are all 0.99 W, corresponding to the pulse peak power of around 300 W. Theoretically, to keep consistent with the analysis shown in Section II-B and quantitatively compare the measurement results, the pulse peak powers of the three systems are supposed to be the same. Then the output power of the noncoding Lidar should be set to near 0.123 W. The issue is that even at the places close to the starting point, the Doppler peaks are not prominent enough to be identified from the noise floor under such output power levels. Therefore, the compared noncoding Lidar with 40 ns pulsewidth has an average output power of 0.32 W with an 800 W pulse peak power. However, the encoded Lidar still performs superior in terms of detection range enlargement and noise suppression than the other two as the results plotted in Fig. 8.

Fig. 8(a) gives a 9-bit coded pulse waveform example from the MZM while Fig. 8(b) shows the beat pulse of the corresponding amplified pulse waveform and the reference beam. The code sequences of the two captured pulses both refer to 101111111. Fig. 8(c) shows the beat pulse of the noncoding Lidar with 360 ns pulsewidth. The waveforms of the three pulses all exhibit a slight shape distortion, where the pulse front edge is higher than the leading edge. The interference pattern in the beat signals of Fig. 8(b) and (c)

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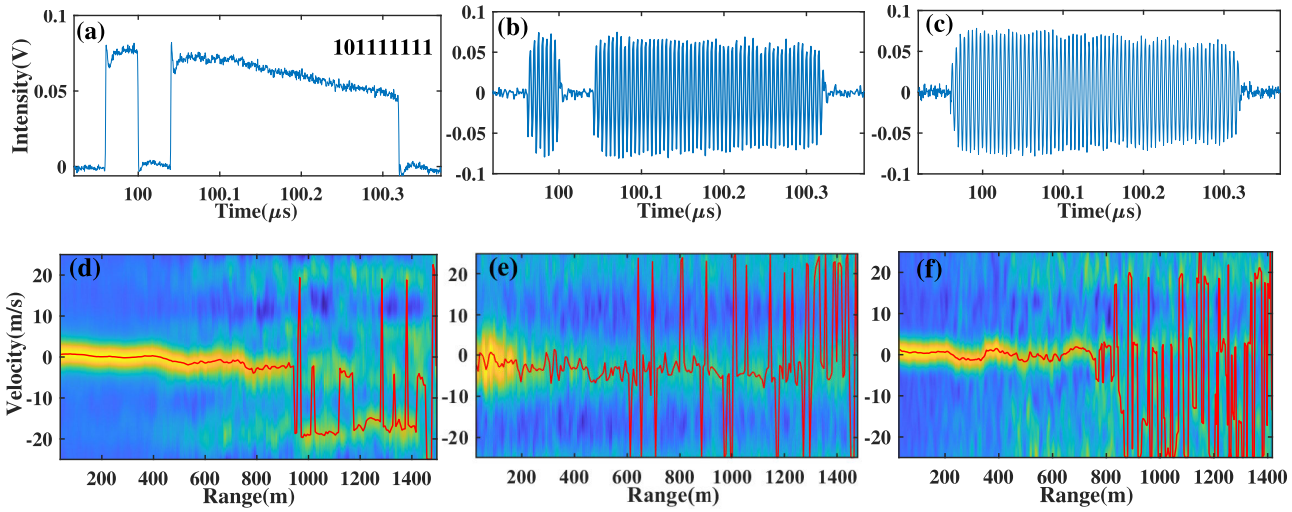


Fig. 8. Measurement results of noncoding Lidars and the 9-bit coding Lidar. (a) Modulated pulse waveform generated by MZM. The beat pulse of the corresponding amplified pulse waveform and the reference beam of (b) 9-bit coding Lidar and (c) noncoding Lidar with 360 ns pulsewidth. The periodogram in the frequency-range diagram and extracted velocities of (d) 9-bit coding Lidar and (e) noncoding Lidar with 40 ns and (f) 360 ns pulsewidths.

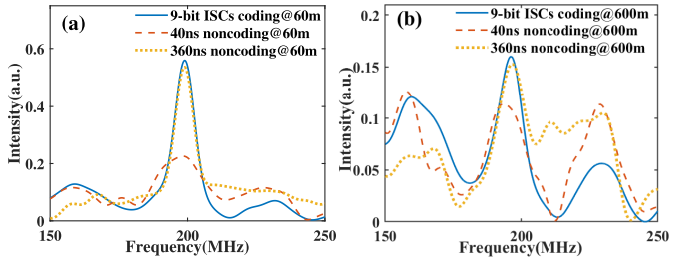


Fig. 9. Power spectra of the 9-bit ISCs coding Lidar and the noncoding Lidars with 40 and 360 ns pulsewidths at distances of (a) 60 and (b) 600 m.

refers to a frequency of 200 M, brought by the AOM frequency shift.

The calculated periodograms of the three systems are shown in Fig. 8(d)–(f) in the frequency-range diagram. Noteworthy is that these results are collected in different moments because of sharing the same hardware. The central red line points out the extracted wind velocities obtained by the spectral peak in each column. Apparently, the velocity trace got from the 9-bit coding Lidar as shown in Fig. 8(d) has the longest smooth part until around 950 m. During the range of 0–400 m, the background color seems pure blue, which means the intensity of the signal peak overwhelms that of the noise. Next, the velocities seem erroneous at the peaks farther than 950 m because the intensity of the noise floor would overtake that of atmospheric peaks at that time.

In contrast, as shown in Fig. 8(e), the velocity trace measured by the noncoding Lidar with 40 ns pulsewidth is the most oscillating. The detectable range of the system is drastically reduced by the onset of sharp peaks that vary up and down earlier at about 600 m. Seen from Fig. 8(f), the noncoding Lidar with 360 ns pulsewidth can provide a reliable detection range of around 800 m, slightly less than the coding Lidar with the same overall pulsewidth. However, the background wind fields are different between the two measurements, which may also influence the presented detection performance. Moreover, the 9-bit ISCs coding Lidar also extends the detection range to around 1.6 times the noncoding Lidar in single-bit duration, similar to the simulations in Section III-B.

Fig. 9 selects the periodograms of the noncoding Lidars and the 9-bit coding Lidar at two different locations. When the sensing range is 60 m, the three atmospheric peaks in Fig. 9(a) are significantly higher than the noise peaks on both sides. However, the spectral peak envelope of the noncoding Lidar with a 40 ns pulsewidth is wider, lower, and more asymmetrical than the other two. This is consistent with the spectral broadening which is inversely proportional to the pulsewidth. When the sensing range enlarges to 600 m, the aerosol signal cannot be distinguished by the noncoding Lidar with a 40 ns pulsewidth, since the background pseudo-peaks have close intensities. However, the aerosol peak continues to dominate all the peaks in the 9-bit coding Lidar, because the correlation-free coding technique can transmit and utilize more energy. For the periodogram of the noncoding Lidar with 360 ns pulsewidth, the signal intensity on both sides close to the atmospheric peak is still high, which makes the peak less prominent than the peak in the periodogram of the coding Lidar. Combining Fig. 9(a) and (b), though the SNR has obviously decreased along with the increasing range, the ISCs coding technique could slow down this trend.

C. Results From 16-Bit, 32-Bit, 64-Bit Coding Lidar With 10 ns Bit Duration

This section aims to discover if the detection distance can be enlarged in accordance with the code lengths when a very small pulsewidth is utilized. The modulation bit duration keeps at 10 ns while the code length is set to 16, 32, and 64. The average output powers of the 16-bit, 32-bit, and 64-bit ISCs coding Lidars are 0.48, 0.98, and 2 W, respectively, in order to maintain a constant pulse peak power. Other parameters are the same as those listed in Table I.

Relative results are displayed in Figs. 10 and 11. Fig. 10(a)–(c) plot the beat signals of the reference beam and the amplified probing pulses which are modulated by the

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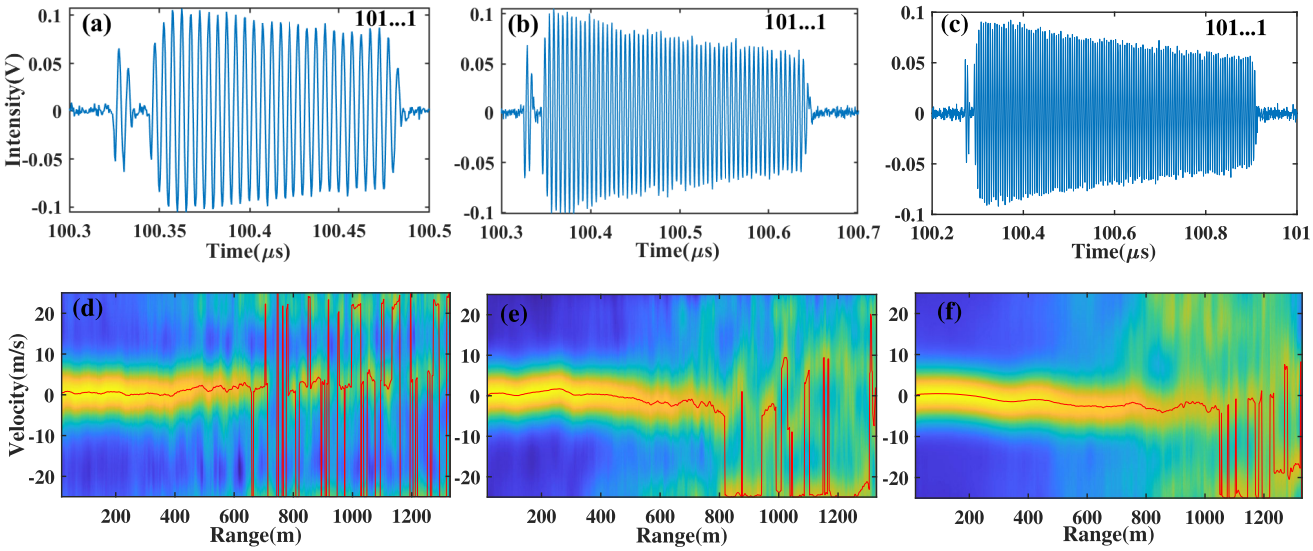


Fig. 10. Beat signals and corresponding periodograms of the 16-bit, 32-bit and 64-bit ISCs coding Lidars. (a)–(c) Beat signals of the amplified pulse waveform and the reference beam of the 16-bit, 32-bit and 64-bit ISCs coding Lidars. (d)–(f) Processed periodograms of the 16-bit, 32-bit and 64-bit ISCs coding Lidars.

second code sequence of the ISCs matrix. At this time, the three probing pulses have different overall widths but all turn off the laser at the second bit. From Fig. 10(a)–(c), along with the increase in overall coding pulsewidth, the deformation of the pulse shape becomes more noticeable. Finally, it is more like a trapezoid than a square. The first split bit maintains around 10 ns, which means that there is no significant time broadening during the experimental modulation process. However, the first bit appears quite lower than the spliced bits in the second part. This is also in line with the fact that a pulse with a very small duration is challenging to be amplified to high peak power. By modifying the variation of the pulse amplitudes into the decoding matrix G_{decode}, the periodogram along the range can be recovered.

Throughout Fig. 10(d)–(f), the reliable detection range continues to extend with the increase in the code length. Overall, at the beginning of the detection, the SNR is high enough to provide credible estimations, so the velocity traces appear smooth with a pure background. However, along with increasing detection range, the spectral noise may exceed the aerosol signal, resulting in inaccurately estimated velocities as the fluctuations in the red solid lines. The farthest detection ranges of the 16-bit, 32-bit, and 64-bit ISCs coding Lidars are around 650, 830, and 1080 m. The maximum detection range of Lidars with the code length of 16, 32, and 64 increases sequentially, which is consistent with the theoretical analysis demonstrated in Section II-B.

Fig. 11 complements the processed spectra of the three systems with different code lengths at distances of 60 and 900 m. The spectra plotted in Fig. 11(a) are normalized in 0–1 for ease of viewing. The intensities of the atmosphere peaks are at 1, while the intensities of the interfering noise decrease with the increase in code length. Hence, the SNR improves with the code length. As seen from Fig. 11(b), when the sensing range enlarges to 900 m, the highest peaks on top of the spectral base of the 16-bit and 32-bit ISCs coding Lidars are noise signals, which would mislead the measurement.

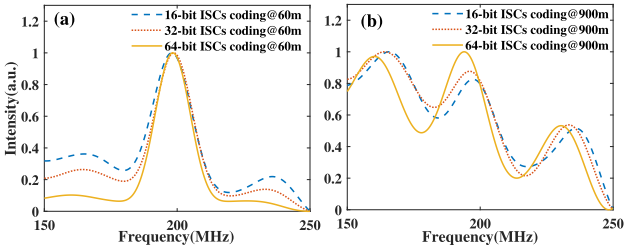


Fig. 11. Periodogram of the 16-bit, 32-bit and 64-bit ISCs coding Lidars at distances of (a) 60 and (b) 900 m.

V. Conclusion

In conclusion, an ISC-based coding Lidar is proposed and constructed, trying to solve the dilemma of spatial resolution and the maximum detection range. The output power/SNR can be enlarged by increasing the applied code length, while the spatial resolution can be improved by narrowing the width of the modulated bit. It is theoretically proved that the ISCs, which we designed, outperform other codes in terms of time and energy efficiency. In simulations, the ISCs coding Lidar has a similar effective detection range but a higher spatial resolution than the noncoding Lidar utilizing an identical overall pulsewidth. The experiment reveals a consistent trend between the calculated range improvement ratio and the real detection range extension of the coding Lidars with various code lengths, demonstrating the viability of the ISC technique in wind Lidar.

However, there are still shortcomings with the ISC technique. The simulated coding Lidar's spatial resolution cannot reach the level of noncoding Lidar with a single-bit duration. Since the decoding method tries to balance the spatial resolution and resultant spectrum power by changing the element y in the decoding matrix, the resolution would also be influenced by the code length. When more codes are implemented, the degradation of resolution enlarges through the accumulation of time-delayed recovered periodograms. To solve this, it is

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necessary and useful to continue studying the best choice of y under different coding settings and detection requirements. Another promising modification direction is to identify the offset ranges and make appropriate compensations. The velocity shift/offset ranges result from the increasing sensing scope brought by larger pulsewidth. If the smoothing effect of a wide sensing scope on velocity estimation can be analyzed, the potential offset range may be determined and appropriate compensations can be made. Moreover, due to the lack of high-resolution reference wind velocities, the accuracy of the constructed coding Lidar cannot be studied. Clearly, additional verification about the actual spatial resolution is warranted. Therefore, efforts will be made to do cross-checking experiments by establishing two high-resolution wind Lidars.

Appendix
Explanation of the Balance of Spatial Resolution and Spectrum Power in Decoding

Based on the power spectra superposition process expressed in (1), there has

$$P_\tau(t) = \left(S_2(t) + S_3(t) + S_4(t) - 2S_1(t)\right)/3. \tag{13}$$

Following (12), the periodogram $P_\tau(t)$ of a single-bit duration τ can be derived. However, after simulations based on this decoding method, the strength of the recovered periodogram $P_\tau(t)$ is low, making it easy to be interrupted by the background noises in velocity estimation, let alone taking advantage of the high single-bit resolution. In order to enhance the strength of $P_\tau(t)$ and realize the potential of high spatial resolution, the decoding matrix can be set as

$$G_{\text{decode}}(4, y) = \begin{bmatrix} -y & 1 & 1 & 1 \\ 1 & -y & 1 & 1 \\ 1 & 1 & -y & 1 \\ 1 & 1 & 1 & -y \end{bmatrix}, \quad 0 < y \le 2. \tag{14}$$

Combining (14), the estimated periodograms of a single-bit duration τ with the time delay $\tau, 2\tau, 3\tau$ can be derived as

$$\begin{aligned} &\begin{bmatrix} \hat{P}_\tau(t) \\ \hat{P}_\tau(t-\tau) \\ \hat{P}_\tau(t-2\tau) \\ \hat{P}_\tau(t-3\tau) \end{bmatrix} \\ &= \begin{bmatrix} P_\tau(t) + (2-y)S_1(t)/3 \\ P_\tau(t-\tau) + (2-y)S_2(t)/3 \\ P_\tau(t-2\tau) + (2-y)S_3(t)/3 \\ P_\tau(t-3\tau) + (2-y)S_4(t)/3 \end{bmatrix} \\ &= G_{\text{decode}}(4, y)/3 \cdot \begin{bmatrix} S_1(t) & S_2(t) & S_3(t) & S_4(t) \end{bmatrix}^T. \end{aligned} \tag{15}$$

When periodograms of different time delays are obtained, the averaged resultant $\hat{P}_\tau(t)$ can be calculated by

$$\hat{\bar{P}}_\tau(t) = \frac{\left[\hat{P}_\tau(t) + \hat{P}'_\tau(t-\tau) + \hat{P}'_\tau(t-2\tau) + \hat{P}'_\tau(t-3\tau)\right]}{4}. \tag{16}$$

When the code length is 4, if $y = 2$, the estimated periodograms $\hat{P}_\tau(t)$, $\hat{P}_\tau(t-\tau)$, $\hat{P}_\tau(t-2\tau)$, and $\hat{P}_\tau(t-3\tau)$ only contain information of the current pulse bit. Thus, $\hat{\bar{P}}_\tau(t)$ has the highest spatial resolution which equals to the single-bit. When the code length is M, y should be set to $M-2$ so that $\hat{\bar{P}}_\tau(t)$ can achieve the resolution of single bit. If $0 < y < M-2$, each estimated periodogram would also contain information from other pulse bits, which means $\hat{\bar{P}}_\tau(t)$ cannot reach the spatial resolution of single-bit but would have higher strength. Higher strength helps to eliminate the interference of noise and fulfill the resolution advantage brought by the narrow bit duration. The drawback is that the resolution of the coded lidar would be limited by both the length of the code and the bit duration. Therefore, it is necessary to choose an optimal y in the decoding matrix in order to balance the spatial resolution and the strength of the power spectrum when decoding.

However, in practical applications, the value of y may be changed to suit different code lengths and bit durations, different wind field detection requirements, and so on. Due to the coupling of various parameters' influences, it is quite complicated to investigate the best choice of y in different cases, which somewhat deviates from the focus of this manuscript. In fact, this article concentrates on putting forward a novel coding scheme to improve the spatial resolution and detection range of CDWL. For ease of observation, the decoding matrix used in the manuscript is set as

$$G_{\text{decode}}(4) = \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}, \quad y = 1. \tag{17}$$

Regardless of the code length M, y is set to 1, which may not be the best choice in all cases but is relatively appropriate. As seen from Figs. 5 and 6, the spatial resolution and detection range of coding Lidar are both improved, which proves the effectiveness of the ISCs coding method and the decoding matrix where $y = 1$ used in this article.

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